

TrES-5b

R-0796

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_220530 · created 2026-07-28T08:21:33+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459729.9136 +/- 0.0036 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.091 +/- 0.034
Transit depth (Rp/Rs)^2	0.82 +/- 0.62 [%]
Orbital Inclination (inc)	84.22 +/- 2.76
Airmass coefficient 1 (a1)	0.937 +/- 0.013
Airmass coefficient 2 (a2)	0.057 +/- 0.091
Scatter in the residuals of the lightcurve fit is	1.44 %
Best Comparison Star	#2 - [223, 398]
Optimal Method	PSF photometry
Transit Duration (day)	0.058 +/- 0.02

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.091 ± 0.034 vs 0.142 (deviation 1.21σ)
Duration fit vs published	0.058 d vs 0.0758 d (deviation 0.72σ)
Mid-time O–C	4.6 min
Depth SNR	2.2
Residual scatter	1.44 %

Light curve

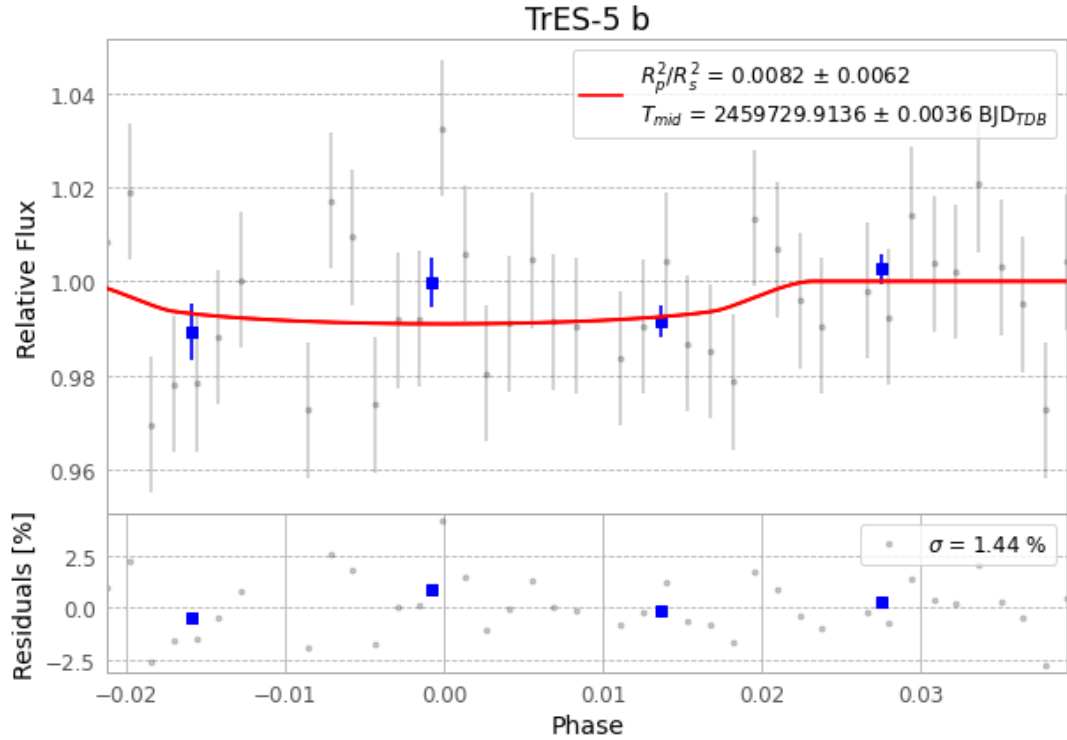


Figure 1 — FinalLightCurve_TrES-5 b_2022-05-30.png (SHA-256 0c9c82cac57cda34...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [362, 363], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 f49dbb6d4338e3d5...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

41 FITS frames (27 MB), TRES-5220530090914.FITS → TRES-5220530111813.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-220530.log (8ab170abbaba...), FinalLightCurve_TrES-5 b_2022-05-30.png (0c9c82cac57c...), FinalLightCurve_TrES-5 b_2022-05-30.csv (b5b8a04745a3...)

Compute resources

Wall time 110.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 08:21Z.